Davide **Aldegheri**

Senior Software Engineer

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Summary

Senior Software Engineer with **10+ years of experience** architecting and developing enterprise-scale desktop applications and interactive touch surfaces. Expert in bridging high-performance graphics rendering with complex backend systems using **.NET, C#, C++, and TypeScript.** Deeply focused on software architecture, real-time data visualization, and delivering seamless User Experiences for large-scale enterprise, governance, and industrial applications.

Skills

Languages & Frameworks	.NET, C#, React/TypeScript, Three.JS, C/C++, Python, JAVA
Core Competencies	Software Architecture, 3D Mathematics, OOP, Programming Patterns, Algorithms, Graphics Programming
Languages	Italian, English

Work Experience

CT Innovation S.r.l.

Verona, Italy

Team Lead & Software Developer — *Apr. 2016 - Today*

- **Technical Leadership:** Directed the Unity3D and React teams in architecting and delivering flagship applications, demonstrating strong adaptability and problem-solving to meet complex client demands.
- **Theater Architecture:** Led R&D and core development for Theater, an advanced multimedia collaboration platform for enterprise, smart cities, and governance, engineered using Unity3D.
- **Trion & Industry 4.0:** Developed immersive real-time visualization features for Trion, using React/Three.js, integrating Real-Time Locating Systems (RTLS) and Ultra-Wide Band (UWB) technology for advanced indoor localization.
- **High-Performance Rendering:** Innovated core functionalities, including a custom Chromium-based browser and WebRTC integration, enabling simultaneous, zero-latency rendering of multi-source web content.
- **Systems Integration:** Architected a suite of C# and C++ plugins and Windows services to bridge Unity3D with native OS features, streamlining database integration and handling highly demanding graphic rendering.

Education

Università degli studi di Verona

Verona, Italy

B.S. in Computer Science and Engineering (Part-time) — *Expected graduation: 2027*

- Pursuing degree part-time to continuously expand technical foundations and theoretical problem-solving.
- **Relevant Coursework:** Computer Graphics, Software Engineering, Programming Challenges

Hackathons & Extracurricular Activity

Game Jams

Online, Worldwide

Game Developer — *2016 - 2020*

- Developed core gameplay, VR interactions with **Oculus**, and VFX on multiple games. Awarded **Best 2D Graphic** in 2016 and **Best Mood Mojo** in 2020.

M.S. course “Programming Challenges”

Verona, Italy

Student | Prof. Romeo Rizzi (Full Mark) — *March 2014 - May 2014*

- My task was to create an algorithm that always makes the best possible move in a non-standard Card Game (Triple Triad - Final Fantasy VIII). This algorithm was then put up against other students’ algorithms in a one versus one scenario. The algorithm implementation was written in **C++** and **ranked 1st.**

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